Preliminary Proposal

Project Title: AI Forensic Triage Tool: Predicting Shooting Incidents in Boston to Prioritize Crime Lab Resources

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Date: March 2026

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1. Introduction

Forensic crime laboratories and police departments process thousands of incidents annually under tight resource constraints. Ballistics analysis, DNA testing, and firearms-related evidence are time-intensive. However, not every reported crime involves a shooting. This project develops an AI Forensic Triage Tool that predicts shooting probability in real time using public Boston data, allowing labs to prioritize high-impact cases and improve public safety outcomes.

2. Problem Statement

Current evidence-processing workflows treat all incidents uniformly, creating backlogs and delaying justice in shooting-related cases. With over 300,000 incidents in the Boston dataset since 2015, manual triage is inefficient. A predictive model using incident features and neighborhood demographics can flag high-probability shooting cases for faster forensic analysis.

3. Data Source and Description

- Boston Police Crime Incident Reports (2023–present): ~150k rows CSV from https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system. Key columns: SHOOTING (binary target), OCCURRED_ON_DATE, DISTRICT, Lat/Long, OFFENSE_CODE_GROUP.
- U.S. Census ACS 2020–2024: Boston neighborhoods/tracts (median income, poverty rate, education, race, housing density) from data.census.gov. These will be merged by district to add socioeconomic context.

Both datasets are public, free, and require original cleaning.

4. Research Question and Hypothesis

Research Question: Can incident features (time of day, location, district, offense proxies) and neighborhood demographics accurately predict whether a reported crime will involve a shooting?

Hypothesis: Nighttime incidents in higher-poverty districts will show ≥35% higher predicted probability of involving a shooting.

5. Variable Justification and Intersections

- Time features (hour_sin/cos, is_night, is_weekend) capture known violent-crime patterns.
- Location (DISTRICT + Lat/Long) reflects Boston's geographic disparities.
- Offense proxies (violent flag) serve as strong predictors.
- Census demographics (poverty, income, education, race) add the social determinants layer. Key interactions: night × high-poverty (expected top SHAP feature).

6. Methodology

6.1 Data Cleaning & Preprocessing: Parse dates, handle missing values, filter invalid coordinates, create binary SHOOTING target.

6.2 Feature Engineering: Circular time encoding, violent proxy, district dummies, census merge, SMOTE for class imbalance.

6.3 Exploratory Data Analysis: Shooting rates by district/hour, correlation matrices.

6.4 Hypothesis Testing: Chi-square tests and preliminary logistic regression.

6.5 Predictive Modeling: XGBoost / Random Forest classifier. Evaluation: Precision-Recall AUC, F1-score. SHAP explainability.

7. Planned Visualization

District heatmaps, time-of-day line plots, SHAP summary and dependence plots, and a final Tableau dashboard for stakeholder presentation.

8. Expected Key Insights

Nighttime and poverty will emerge as dominant predictors. The model is expected to achieve >0.80 PR-AUC with clear, actionable SHAP explanations for forensic triage.

9. Target Audience/Stakeholder

Massachusetts State Police Crime Laboratory and Boston Police Department, direct users for real-world case prioritization and resource allocation.

10. Conclusion

This Boston-specific, public-data project delivers a practical AI tool with immediate public-safety value.

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